

Adwait Hegde

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EDUCATION

Texas A&M University · College Station, TX
(Incoming) **M.S. Computer Science**

Expected: May 2028

Sardar Patel Institute of Technology · Mumbai, IN
B.Tech in Computer Engineering

May 2023

GPA: 9.46/10 (3.89/4) **Minors:** Introduction to Management (SPJIMR)

Relevant Coursework: AI & Machine Learning · Deep Learning · NLP · Soft Computing · Big Data Analytics · Data Structures · Design & Analysis of Algorithms · Operating Systems · Software Engineering · Compiler Construction · Theory of Computation · Discrete Structures

TEACHING EXPERIENCE

Sardar Patel Institute of Technology · Mumbai, IN

Jan 2023 – Jun 2023

Teaching Assistant — AS202: Python Programming for Data Science Lab

- Authored and benchmarked custom **HackerRank** challenges focused on *NumPy*, *Pandas*, and *Scikit-learn* to assess 120+ students' proficiency in data preprocessing and model evaluation.
- Spearheaded the **grading workflows automation** by implementing custom test cases, achieving 90% reduction in evaluation latency.
- Facilitated weekly technical recitations and **1:1 code reviews**, mentoring students on algorithmic efficiency, resulting in an increase in average lab assessment scores.
- Curated a comprehensive **Python for Data Science**, including modular tutorials on complexity analysis, OOP principles, and supervised learning frameworks to supplement core course lectures.

Igniting Insight · NGO

Sep 2020 – Sep 2022

Igniter — Mathematics Faculty (2 years · 50+ lecture hours)

- Delivered **Mathematics** instruction (Std X & XII) to students from underserved urban communities, covering full board syllabi and enabling improved conceptual clarity and exam readiness.
- Designed and facilitated a structured **Logical Reasoning workshop** for secondary students, strengthening analytical thinking and problem-solving frameworks.
- Adapted teaching methods to **diverse learning needs**, improving student engagement.
- Mentored students beyond curriculum, fostering confidence and independent learning habits.

Abhyudaya – SPJIMR · Non-classroom Learning Initiative, SPIT

Sep 2022 – Dec 2022

Student Volunteer — Teaching Assistant (Mathematics & Science)

- Tutored secondary-school students from underprivileged urban communities in Mathematics and Science, simplifying complex concepts and improving their academic confidence and performance.
- Contributed to SPIT's non-classroom learning initiative by designing engaging learning activities and providing personalized support.
- Improved **administrative efficiency** by overseeing student onboarding, data entry, and documentation.

RESEARCH PUBLICATIONS

Fall Detection System via Computer Vision · ICACCS 2023 — IEEE

[\[IEEE Link\]](#)

- Software-centric approach eliminating hardware dependency by detecting falls through analysis of **center-of-gravity shifts**, inclination angle, and velocity of descent.

Intelligent Traffic Management System · INDIACom 2023 — IEEE

[\[IEEE Link\]](#)

- Fuzzy Logic Controller + YOLO v3** object detection to optimize signal durations; CNN for emergency vehicle identification — reduced average wait time by ~20%.

ML Models for Fantasy Premier League Prediction · ASIANCON 2022 — IEEE

[\[IEEE Link\]](#)

- Comparative evaluation of Linear Regression, Decision Trees, and Random Forests over a custom feature-engineering pipeline.

INDUSTRY EXPERIENCE

Barclays · Software Engineer — Pune, IN

Jul 2023 – Present

- Wrote and tuned complex SQL queries against PostgreSQL for the Strategic Debt Manager (SaaS) platform, applying indexing and query-plan optimization to reduce batch processing time by 20%.
- Designed relational schema logic and SQL-backed data access layers powering RESTful APIs for Collections & Recoveries (Amazon Barclaycard), built with multi-threading and Java Stream.
- Built and maintained SQL-driven ETL data pipelines, and automated Jenkins CI/CD pipelines, cutting deployment time by 90%.

SKILLS & ACHIEVEMENTS

Core Areas: Computer Vision, Deep Learning, NLP, Soft Computing, Data Structures & Algorithms

Programming: Python, Java, SQL, C, C++

Tools: Spring Boot, Django, Flask, PostgreSQL, MongoDB, Docker, Jenkins, Elasticsearch, Git

Hackathons: CCT Docker Hackathon @ Barclays (Winner, 20 teams) · BUK Code-A-Thon (Finalist, 65+ teams) · AthenaHack 2.0 (2nd Runner-up) · CODATRON++ (1st Runner-up, 150+ participants)